

E-cigarette Supply Side

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2026-2027 academic year

0.1 Marc Meeting August 8, 2026

- Demand
 - Nested logit: ecigarette in one branch, outside option in another branch.
 - * Mean utility = quality (integer)
 - * Do this every period because demand is static
 - construct state space from mean utility
 - Each firm makes choice to invest or not. When you enter, pay a fixed cost. If you invest, this is a forward-looking decision
 - Suppose BLU did not invest, and others did. Mystic was a potential entrant in 2013. In period 2, first centurys smoke exited.
 - * If I invest, they know if other firms invest, I'll leave

0.2 Marc Meeting August 10, 2026

Following the August 5 meeting, I met with Ariel Pakes, and he recommended EBE, not MPE, due to high state space dimensionality.

- Measure of quality
 - See it as ξ in BLP (unobserved quality)
 - How much production you've done $+\xi$, where how much production you've done is "level of accumulated quality" and ξ evolves exogenously
 - You pick your quantity = output produced. Quality affects price.
 - Static nested logit
 - As you pick more quality and it decreases costs
 - "Learn and forget"
- If multiple quality levels
 - ξ cost of nicotine delivery

- * Which do you have to invest in
 - * Endogeneous in that investment does not affect it
 - * ξ exogeneous via markov. just to match the data
 - * Need to track ξ of rivals
- Max of 5 firms per period
- Everybody can choose to exit
- Only one potential entrant
- every set of parameters, solve Bellman and best response
 - Conditional on Bellman, iterate on best response
 - Conditional on best response, update value function
 - Update value function, update best response functions
 - Update best response, update value function
- Pakes and McGuire solve this all
 - Draw choices for next period from strategies
 - Iterate like that . Focus updating of value function on states that firms go to
- Cost of nicotine per unit. Maybe JUUL invested in cost of nicotine
- Does nicotine decrease when the FDA makes changes?
 - Ex. of change is FDA deeming rule, regulating nicotine salts, reprimanding JUUL, etc
 - If so, put nicotine in as a cost
 - Nicotine per product by firm : N_{hom} or mg/mL
 - Do not do by sales-weight. Maybe do by product offering set (number of UPCs)
- Write down a model
 - Investment in cost or investment in ξ
 - Is it draws?
 - Ignore product type
 - 5 firms $\times$ 1 endogenous quality measure $\times$ 1 ξ measure
 - Benkard
 - Ericson Pakes does continuous investment $\rightarrow$ Probability(draw) $\rightarrow$ 0 or 1 binary investment
- For upcoming internal conference:

- Do cost and quality as exogenous, choose static price, leave quantity endogenous
- Firms pick entry, exit, price. This will be dynamic
- There will be cap in market